

Experiment 5.2: Design and Analysis of an FIR High-Pass Filter Using the Hamming Window Technique in MATLAB

Aim: To design and analyze an FIR High-Pass Filter using the Hamming Window technique in MATLAB.

MATLAB Code

```
clc;
clear;
close all;

%% Save Command Window Output
diary('Exp5_2_Output.txt');

%% --- Input Parameters ---
fp = 1000;      % High-pass cutoff frequency (Hz)
fs = 8000;      % Sampling frequency (Hz)
N = 50;         % Filter order

wc = fp/(fs/2); % Normalized cutoff (0-1)

%% --- FIR High Pass using Hamming Window ---
b_hamm = fir1(N, wc, 'high', hamming(N+1));

disp('--- High-Pass Coefficients (Hamming Window) ---');
disp(b_hamm);

%% --- Frequency Response Plot ---
figure(1);
freqz(b_hamm,1);
title('High-pass FIR using Hamming Window');

%% Stop Saving Command Window
diary off;

%% Save Figure (300 DPI)
exportgraphics(figure(1), 'Exp5_2_Hamming_Window.png', 'Resolution', 300);

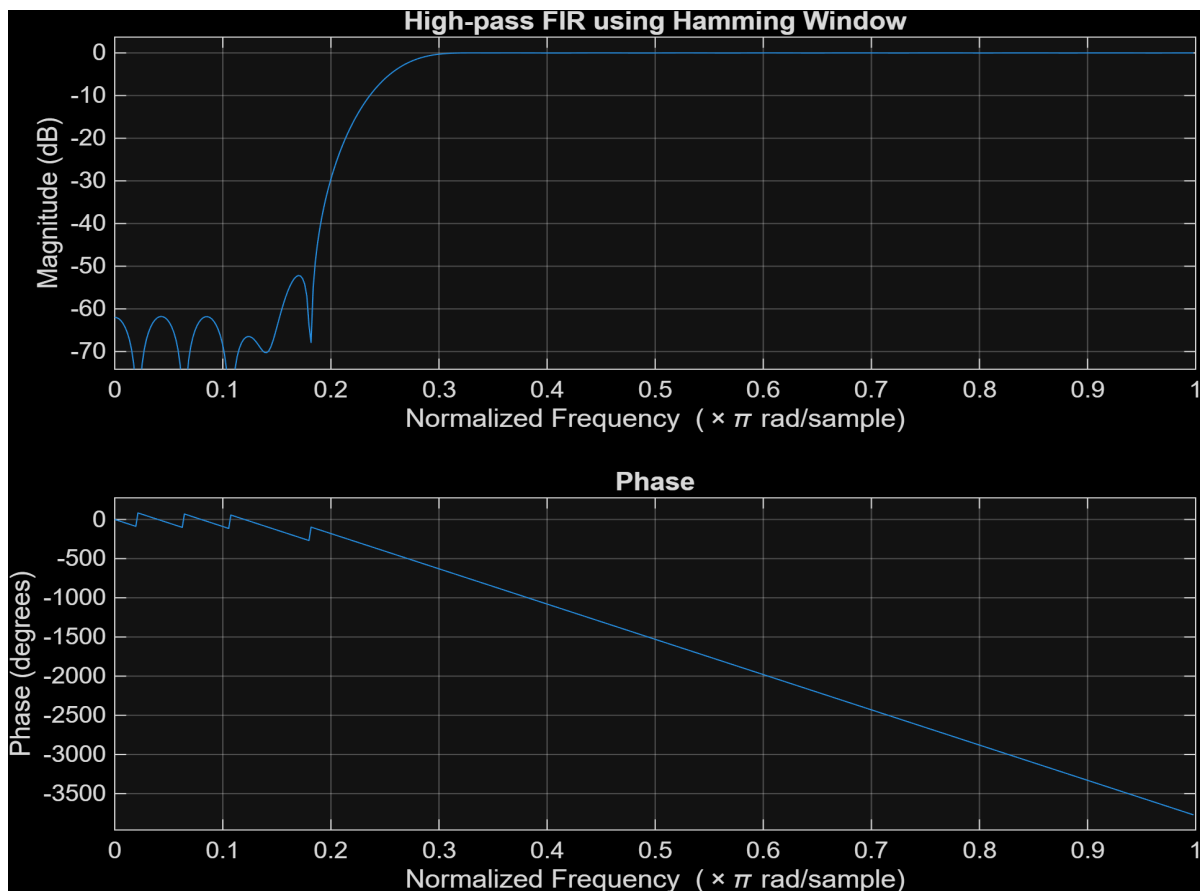
%% Completion Message
disp(' ');
disp('=====');
disp('Experiment 5.2 Completed Successfully');
disp('Files Saved:');
disp('Exp5_2_Output.txt');
disp('Exp5_2_Hamming_Window.png');
disp('=====');
```

Output

```
--- High-Pass Coefficients (Hamming Window) ---
-0.0007
  0
  0.0009
  0.0016
  0.0015
  0
-0.0024
-0.0044
-0.0039
  0
  0.0060
  0.0103
  0.0088
  0
-0.0128
-0.0217
-0.0184
```

0
0.0268
0.0464
0.0410
0
-0.0725
-0.1567
-0.2240
0.7493
-0.2240
-0.1567
-0.0725
0
0.0410
0.0464
0.0268
0
-0.0184
-0.0217
-0.0128
0
0.0088
0.0103
0.0060
0
-0.0039
-0.0044
-0.0024
0
0.0015
0.0016
0.0009
0
-0.0007

Output Image



Result: The FIR High-Pass filter using the Hamming Window technique was successfully designed and analyzed in MATLAB. The frequency response and phase response were obtained successfully.